

Effects of Personalized Generative AI on Student Creativity: Evidence from Creative Performance, Self-efficacy, and Attention Allocation

Yixuan Chen[1], Xuemei Wu[1(✉)], Yihua Zhong[2], and Changqin Huang[1,3(✉)]

[1] Zhejiang Key Laboratory of Intelligent Education Technology and Application, Zhejiang Normal University, No. 688 Yingbin Avenue, Jinhua 321004, Zhejiang, China
wuxuemei@zjnu.edu.cn, cqhuang@zju.edu.cn

[2] Shanghai Institute of Artificial Intelligence for Education, East China Normal University, Shanghai 200062, China

[3] School of Education, Zhejiang University, No. 866 Yuhangtang Road, Hangzhou 310058, Zhejiang, China

Abstract. Personalized generative AI (GAI) holds promise in supporting student creativity, though evidence regarding its effectiveness remains limited. This study employed a within-subject design to compare the effects of standard versus personalized GAI on students' creative performance, self-efficacy, and attention allocation. Each of the 62 students engaged in two creativity tasks: one with standard GAI (ChatGPT-4) providing generic feedback, and the other with a personalized GAI tutor (Mr. Ranedeer), implemented via prompt engineering to deliver goal-aligned feedback tailored to students' needs. Results indicated that personalized GAI improved creative performance in output quality and elaboration, boosted self-efficacy, and increased gaze transitions across task-relevant areas of interest (AOIs) as evidence of attention allocation. Furthermore, higher self-efficacy was associated with more transitions across these areas. These findings contribute to understanding the mechanisms through which personalized GAI influences student creativity, providing valuable insights into the design of strategies that promote GAI-supported creative development.

Keywords: generative artificial intelligence · personalized learning · creativity · creative self-efficacy · eye-tracking

1 Introduction

Creativity is widely recognized as a fundamental capacity for addressing complex, real-world challenges and serves as a central objective in contemporary educational initiatives (Beghetto and Kaufman, 2014; Runco and Jaeger, 2012). The rapid development of GAI, particularly the integration of large language models (LLMs) into education, has drawn attention for augmenting human creativity (Lee and Chung, 2024). General-purpose GAI tools, such as ChatGPT, can stimulate students' creativity through real-time feedback

© The Author(s), under exclusive license to Springer Nature Singapore Pte Ltd. 2026
S. K. S. Cheung et al. (Eds.): ICTE 2025, CCIS 2752, pp. 376–388, 2026.
https://doi.org/10.1007/978-981-95-4499-8_31

and ideational prompts (Habib et al., 2024). However, existing systems lack the ability to perceive learner states dynamically or adapt to task progression. Consequently, ChatGPT often fails to align with individual differences or task demands, limiting its effectiveness in fostering sustained engagement (Lo et al., 2024).

Personalized GAI has the potential to address these limitations by delivering context-sensitive feedback tailored to individual learners' cognitive and emotional states (Guettala et al., 2024). This approach not only marks a significant advancement in autonomous learning but also represents a transformative shift in how student creativity is cultivated. Despite its promising potential, however, a critical gap remains in the empirical understanding of its impact. Therefore, it is of particular importance to examine its actual effects on student creativity.

Creativity arises not solely from technological affordances but through the dynamic interaction between learners and their environment (Sternberg and Lubart, 1995). In human–GAI collaboration, two learner-related factors are crucial. First, creative self-efficacy, defined as individuals' belief in their ability to generate original and useful ideas, shapes how students engage with creative tasks (Tierney and Farmer, 2002). While ChatGPT has been shown to enhance self-efficacy, such gains do not always improve performance (Urban et al., 2024). In contrast, personalized GAI may better regulate self-efficacy by offering context-sensitive and socially persuasive feedback, fostering more strategic and sustained engagement.

Second, attention allocation plays a central role in creative cognition. According to the generative–exploratory model, creative thinking relies on the dynamic coordination between external input and internal ideation (Finke and Smith, 1992). In GAI-supported contexts, students must continually interpret AI feedback while developing their own ideas, making attention allocation a key cognitive mechanism underlying human–GAI co-creation. Despite the importance of both factors, how they are shaped by different GAI types and link to students' creativity remains underexplored.

To address this gap, the present study employed a within-subjects design to compare the effects of personalized and standard GAI on students' creative performance, self-efficacy, and attention allocation. By utilizing multi-modal data, this research not only sought to provide empirical evidence on the effectiveness of personalized GAI in fostering student creativity, but also offering novel insights into the cognitive and motivational aspects underpinning human-GAI co-creation.

2 Literature Review

2.1 Generative AI in Fostering Student Creativity

Creativity is commonly defined as the ability to generate ideas that are both novel and useful (Runco and Jaeger, 2012). In educational contexts, creativity is increasingly recognized as essential for promoting deep understanding and interdisciplinary learning (Hernández-Torrano and Ibrayeva, 2020).

Generative AI has emerged as a powerful tool for fostering student creativity by providing real-time feedback and supporting divergent thinking through high-quality formative feedback (Lee and Chung, 2024). Empirical evidence shows that general-purpose GAI tools, such as ChatGPT, can enhance students' creative problem-solving

(Urban et al., 2024). However, several limitations have been identified. First, the feedback often lacks personalization, failing to address learners' diverse needs and potentially reducing engagement (Mogavi et al., 2024). Second, students frequently interact with ChatGPT passively, accepting suggestions without integrating them into their own thinking (Lo et al., 2024). Finally, under high cognitive load, current AI systems offer limited emotional responsiveness, which may hinder motivation and sustained engagement (Joksimovic et al., 2023).

Personalized GAI holds promise in addressing these challenges. According to Yeh (2011), personal factors are central to creativity development. In computer-based environments, personalized learning has shown potential to enhance students' creative performance and motivation through adaptive learning paths (Lin et al., 2013). Extending this line of inquiry into the GAI era, personalized GAI systems can dynamically adapt feedback strategies based on individual characteristics and task progression, offering more targeted cognitive and emotional support (Maity and Deroy, 2024). Drawing on the Computers as Social Actors framework (Reeves and Nass, 1996), AI agents with human-like traits can increase perceived social presence and foster students' willingness to collaborate (Skulmowski, 2024). However, excessive reliance on such support may reduce students' autonomy in generating original ideas (Habib et al., 2024). Despite the promising potential of personalized GAI, empirical studies on its direct impact remain limited and inconsistent, underscoring the need for further research to clarify its true benefits and underlying mechanisms.

2.2 Creative Self-Efficacy and Its Relation with Creativity

Creative self-efficacy refers to an individual's confidence in their ability to generate creative outcomes (Tierney and Farmer, 2002). It serves as a key motivational resource that sustains effort and encourages exploration in a creative context (Puozzo and Audrin, 2021). However, excessively high self-efficacy may lead to inaccurate task evaluation and reduced effort investment (Vancouver and Kendall, 2006).

In GAI-supported creative learning, creative self-efficacy plays a pivotal role in shaping how learners engage with and process AI feedback. Although recent studies suggest that ChatGPT can boost students' creative self-efficacy, these increases have not been consistently linked to significant improvements in creative gains (Urban et al., 2024). One possible explanation is that they may overestimate their abilities, leading to reduced effort in refining their work. Previous research has identified performance expectancy, facilitating conditions, and social influence as key factors in boosting self-efficacy (Wei et al., 2022). Building on this, personalized GAI may more effectively foster self-efficacy by adapting feedback to students' cognitive and emotional states during the creative process. However, the specific effects of personalized GAI on creative self-efficacy and its relation with creative performance remain underexplored.

2.3 Eye-Tracking and Attention Allocation in AI-Supported Creative Tasks

Attentional processing is a core mechanism underlying creativity. In collaborative settings, the allocation of attention influences how individuals select, organize, and integrate

information, ultimately shaping the quality of their creative ideas (Di Vincenzo et al., 2021).

Eye-tracking has been widely used in education and human–computer interaction research to investigate attentional patterns. It captures learners' gaze locations and durations in real time, offering objective indicators for examining cognitive processes (Liu and Cui, 2025). In online collaborative settings, attention to peers' ideas, as indicated by mean fixation duration, has been shown to predict creative performance (Pi et al., 2025). Frequent gaze transitions indicate the integration of multiple information sources, whereas longer mean fixation durations suggest deeper processing and sustained attention (Coskun and Cagiltay, 2022).

As a human-like collaborator (Lo et al., 2024), GAI provides continuous feedback throughout the learning process. For students to fully benefit from this feedback, they must interpret, evaluate, and integrate it into their work. How students allocate their attention to AI-generated feedback may reveal their information processing strategies and cognitive effort distribution, ultimately influencing creativity. While eye-tracking technology has proven effective in uncovering cognitive processes, empirical studies on how GAI feedback affects attention allocation during creative tasks remain limited, particularly in comparing different feedback mechanisms.

2.4 The Present Study

Building on the gaps identified in the literature review, this study employed a within-subject experimental design to examine how personalized GAI impacts students' creative performance, self-efficacy, and attention allocation, comparing these effects to standard GAI. To better understand the underlying mechanisms of creativity, the interrelationships among variables are examined from an exploratory perspective. Therefore, the study aims to address the following research questions:

(1) Does personalized GAI enhance students' creative performance and self-efficacy compared with standard GAI?
(2) How does students' attention allocation during creative tasks differ between personalized and standard GAI conditions?
(3) How do the interrelationships among creative performance, creative self-efficacy, and attention allocation vary between personalized and standard GAI conditions?

3 Method

3.1 Participants

Sixty-two students (32 undergraduates and 30 graduate students; 33.9% male) participated in the study. Their ages ranged from 18 to 27 years ($M = 22.6$, $SD = 2.5$), and their academic backgrounds spanned social sciences, life sciences, and technical disciplines. Almost all participants reported prior experience with GAI tools, with greater familiarity reported for ChatGPT than for Mr. Ranedeer. All participants provided informed consent and received compensation for their participation. Ethical approval was granted by the local ethics committee.

3.2 Apparatus

3.2.1 GAI Systems

Two GAI systems were used: a standard GAI (ChatGPT-4) and a personalized GAI (Mr. Ranedeer), both built on OpenAI's GPT-4 model and capable of responding to student prompts in natural language. In the standard condition, ChatGPT-4 provided generic feedback through two stages (Fig. 1, top): offering broad suggestions in response to students' queries, followed by solution generation upon further input. This process supported open-ended exploration with minimal structure. In contrast, the personalized condition employed Mr. Ranedeer, a GPT-4-powered tutor customized via OpenAI's "Explore GPTs" platform. The interaction unfolded in three stages (Fig. 1, bottom), grounded in principles of personalized learning by adapting personalized feedback to enhance creativity (Hattie and Timperley, 2007; Yeh, 2011). First, students configured their learning preferences (e.g., feedback tone, cognitive style) and co-designed a personalized learning plan aligned with their goals, reflecting the importance of tailoring learning experiences to enhance motivation and engagement. Second, in the heuristic questioning phase, Mr. Ranedeer used Socratic-style prompts to encourage reflection and guide exploration, a method consistent with personalized approaches that stimulate deeper thinking. Finally, in solution elaboration, the agent provided goal-aligned feedback and strategic prompts (e.g., idea refinement, elaboration cues), reinforcing the role of personalized feedback in fostering creativity.

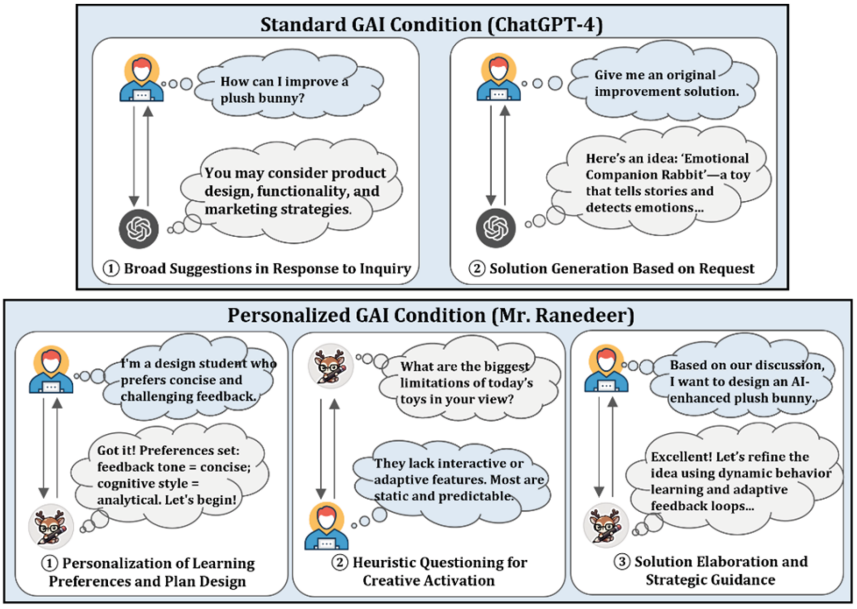


Fig. 1. Interaction stages across standard and personalized GAI conditions.

3.2.2 Eye-Tracking

Eye movements were recorded using a Tobii TX300 Pro Spectrum eye tracker with a 300 Hz sampling rate. A 9-point calibration and validation procedure was conducted before task execution to ensure accurate tracking. Participants completed the creative tasks in a screen-based setup with a flat-panel LCD monitor positioned 60 cm away.

3.3 Procedure

Figure 2 illustrates the experimental procedure. This study was conducted in a quiet laboratory with participants tested individually. After a demographic questionnaire, they received instructions for using ChatGPT-4 and Mr. Ranedeer. A counterbalanced design was used: half completed the first task with ChatGPT-4 and the second with Mr. Ranedeer, while the other half followed the reverse order. Before each task, participants filled out a creative self-efficacy questionnaire for the assigned GAI, then performed the task via the OpenAI web interface. A five-minute break separated the tasks. The experiment lasted about 17 min (SD = 7) in the standard condition and 20 min (SD = 6) in the personalized condition.

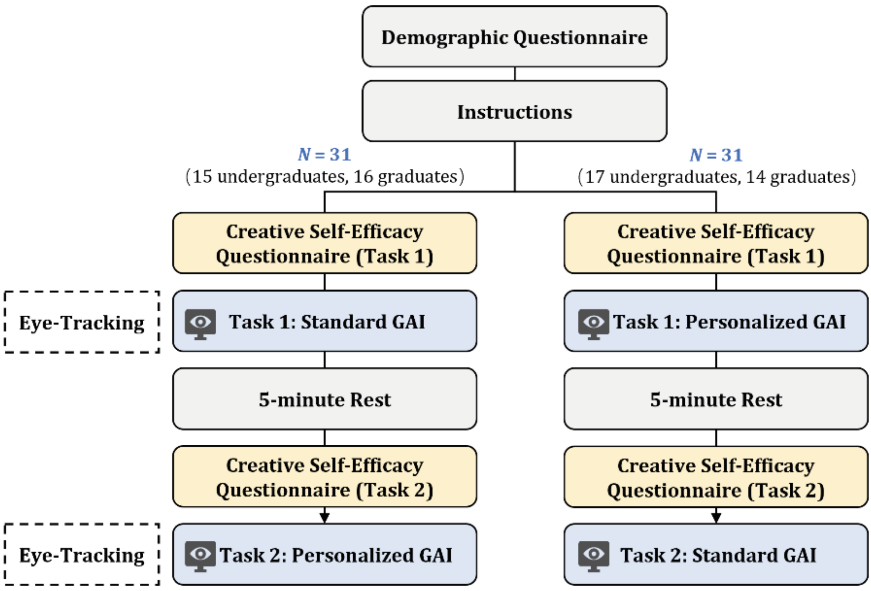


Fig. 2. Experimental procedure in the within-subject design.

3.4 Measures

3.4.1 Creative Performance

Creative performance was measured using the Product Improvement Task (Torrance, 1966). In this task, participants were asked to propose three solutions as original and useful as possible for two common products (e.g., a stuffed bunny and a portable coffee

machine) using the assigned GAI tool. Performance was evaluated on three dimensions: quality, elaboration, and originality. They were rated on a five-point Likert scale (1 = worst, 5 = best) by two trained raters blinded to the experimental conditions. Quality refers to the extent to which a solution addresses the task objective, elaboration to the amount of detail provided, and originality to the novelty of the idea. Solutions were presented in randomized order, and inter-rater reliability was high (Cronbach's α = 0.88, 0.90, 0.86, respectively). Final scores were calculated as the average of the two raters' evaluations.

3.4.2 Creative Self-Efficacy

Creative self-efficacy was assessed using a scale adapted from the instrument developed by Urban et al. (2024). Prior to each task, participants completed a 7-point Likert scale (1 = strongly disagree, 7 = strongly agree) assessing their perceived ability to generate original and useful ideas with the GAI system assigned for that task. Each scale comprised four statements (e.g., "With the help of [the assigned GAI], I am confident that I can come up with three original and useful ideas to improve the product"), with [the assigned GAI] referring to either ChatGPT-4 or Mr. Ranedeer depending on the experimental condition. Internal consistency was high in both conditions (Cronbach's $\alpha = 0.89, 0.88$, respectively).

3.4.3 Eye-Tracking Metrics

To examine students' attention allocation during the creative tasks, three areas of interest (AOIs) were defined in the split-screen layout (see Fig. 3): (a) AI feedback (AOI 1), (b) user input (AOI 2), and (c) workspace (AOI 3). Two eye-tracking metrics were employed. The first, mean fixation duration, was calculated as the total fixation time divided by the number of fixations within each AOI, indicating processing depth; longer fixations reflect greater attentional engagement (Pi et al., 2025). The second, number of transitions, referred to gaze shifts between AOIs, indicating active integration of information (Coskun and Cagiltay, 2022). Only AOI 1 and 3 were analyzed, as they reflected engagement with AI feedback and the externalization of creative ideas, respectively. AOI 2 was excluded because it served solely as a typing interface.

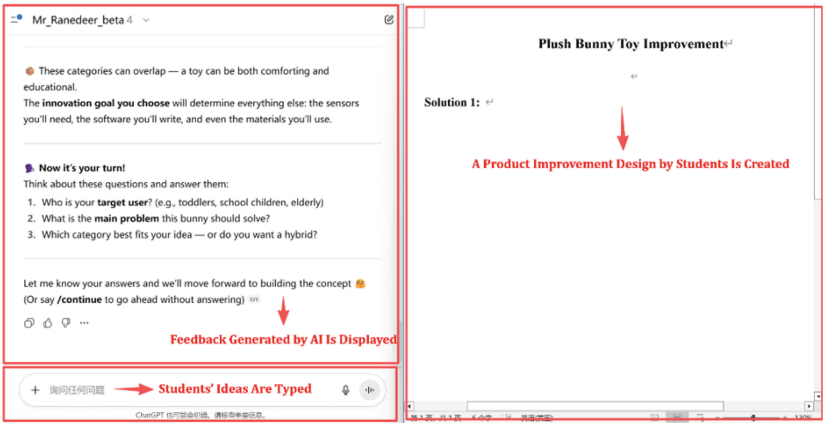


Fig. 3. The split-screen layout consists of three functional areas.

3.5 Data Analysis

Our data comprised students' creative performance (quality, elaboration, originality), creative self-efficacy, and attention allocation (mean fixation duration, number of transitions) in both GAI conditions. Differences between conditions were examined using Wilcoxon Signed-Rank tests, as the Shapiro-Wilk test ($p < 0.05$) indicated violations of normality for all measures. To explore how interrelationships among creative performance, attention allocation, and creative self-efficacy varied across conditions, exploratory correlational analyses were conducted within each condition.

4 Results

4.1 Creative Performance and Self-Efficacy Across GAI Conditions (RQ1)

Given that the data violated normality assumptions, Wilcoxon Signed-Rank tests were conducted on both creative performance and self-efficacy. As shown in Fig. 4, students scored significantly higher in the personalized condition for quality (M = 3.50, SD = 1.00) than in the standard condition (M = 3.08, SD = 1.12; $p = .001$; $r = .42$). Elaboration scores were also significantly higher in the personalized condition (M = 3.58, SD = 1.17) compared with the standard condition (M = 2.48, SD = 1.04; $p < .001$; $r = .62$). No significant difference was observed for originality, with scores slightly lower in the personalized condition (M = 2.92, SD = 1.15) than in the standard condition (M = 3.13, SD = 1.27; $p = .098$; $r = .18$). As for creative self-efficacy, ratings were significantly higher in the personalized condition (M = 5.24, SD = 0.99) than in the standard condition (M = 4.89, SD = 1.19; $p = .033$; $r = .26$). These results suggest that personalized GAI enhanced students' ability to produce more detailed and higher-quality ideas, and strengthened their confidence in their own creative abilities, while showing no clear advantage for originality.

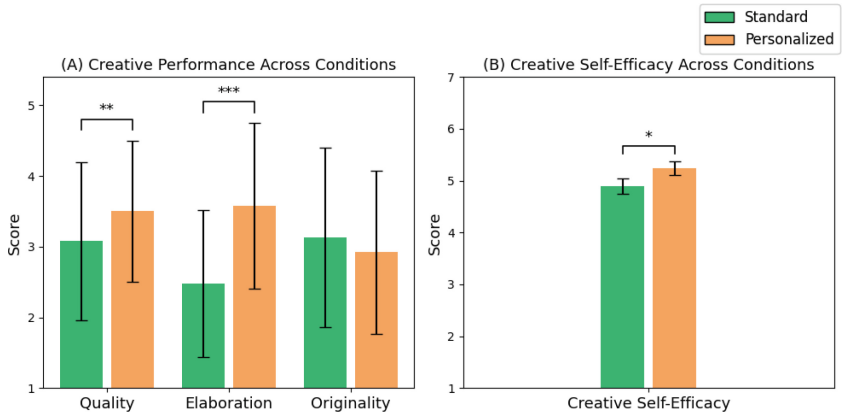


Fig. 4. Comparisons of (A) creative performance and (B) creative self-efficacy across GAI conditions. Error bars indicate standard deviations. *$p < .05$, **$p < .01$, ***$p < .001$.

4.2 Attention Allocation Across GAI Conditions (RQ2)

Wilcoxon Signed-Rank tests were conducted on attention allocation metrics. As shown in Table 1, mean fixation duration in the AI feedback area (AOI 1) and the workspace (AOI 3) did not significantly differ between conditions ($p = .465$ and $p = .756$), suggesting comparable levels of attentional engagement in each region. However, the number of transitions between AOIs was significantly higher in the personalized condition ($p = .017$, medium effect size). Students made 40.71 more transitions when using the personalized GAI (M = 219.37, SD = 13.05) compared to the standard GAI (M = 178.66, SD = 12.83). This increase suggests more frequent gaze shifts between external feedback and idea elaboration (as represented by AOI 1 and AOI 3). Overall, these findings indicate that personalized GAI may have facilitated more dynamic coordination between information sources during the creative task, potentially supporting more flexible cognitive processing.

Table 1. Descriptive statistics and Wilcoxon Signed-Rank test results of attention allocation.

Eye-tracking metrics	Standard Mean (SD)	Personalized Mean (SD)	Z	*p*	Eff. *r*
Mean fixation duration in AOI 1	233.06 (4.24)	230.48 (4.42)	-0.73	0.465	0.09
Mean fixation duration in AOI 3	247.82 (5.54)	246.45 (4.87)	-0.31	0.756	0.04
Number of transitions	178.66 (12.83)	219.37 (13.05)	-2.39	0.017*	0.30

Notes. Eff. r refers to Cohen's r as a measure of effect size, with r = 0.1, 0.3, and 0.5 indicate small, medium, and large effects, respectively. * p < .05, ** p < .01, *** p < .001

4.3 Relationships Among Outcome Variables Across GAI Conditions (RQ3)

To investigate the associations among outcome variables in different GAI conditions, Pearson correlation analyses were conducted separately for the personalized (upper triangle) and standard (lower triangle) conditions.

Table 2. Correlations among outcome variables in the personalized *(upper-triangle)* and standard *(lower-triangle)* GAI conditions.

Outcome variables	1.	2.	3.	4.	5.	6.	7.
1.Creative performance (quality)	–	0.713**	0.746**	-0.143	-0.230	0.110	-0.019
2.Creative performance (elaboration)	0.672**	–	0.634**	-0.080	-0.133	0.203	0.028
3.Creative performance (originality)	0.828**	0.491**	–	-0.081	-0.178	0.071	0.010
4.Mean fixation duration in AOI 1	-0.116	-0.280*	0.103	–	0.704**	-0.082	-0.139
5.Mean fixation duration in AOI 3	-0.159	-0.221	0.096	0.803**	–	-0.016	0.000
6.Number of transitions	-0.032	0.034	-0.043	0.284*	0.261*	–	0.485**
7.Creative self-efficacy	-0.126	-0.092	0.023	0.317*	0.266*	0.468**	–

As presented in Table 2, several consistent patterns were observed. First, the three dimensions of creative performance were highly interrelated: quality correlated with elaboration ($r = .713$; $r = .672$) and originality ($r = .746$; $r = .828$), while elaboration correlated with originality ($r = .634$; $r = .491$). Second, mean fixation duration in AOI 1 and AOI 3 was strongly related ($r = .704$ and $r = .803$), indicating a stable attentional pattern across GAI tools. Third, the number of transitions was positively associated with creative self-efficacy ($r = .485$ and $r = .468$), suggesting that students with higher self-confidence switched actively between areas of AI feedback and idea development.

Differences emerged only in the standard condition. First, mean fixation duration in AOI 1 was negatively associated with elaboration ($r = –.280$), suggesting that longer fixation on AI feedback tended to accompany less elaborated ideas. Additionally, the number of transitions was positively associated with fixation duration in AOI 1 and AOI 3 ($r = .284$ and $r = .261$). Creative self-efficacy was also positively associated with fixation duration in these regions ($r = .317$ and $r = .266$). Taken together, these findings suggest that with general-purpose GAI, students with higher self-efficacy may engage in deeper cognitive processing and more active integration of external feedback with internal idea development. This pattern did not emerge with personalized GAI.

5 Discussion

This study compared personalized and standard GAI in terms of creative performance, creative self-efficacy, and attention allocation, offering new insights into how personalization shapes human–GAI co-creativity. Three major findings emerged.

First, personalized GAI improved the quality and elaboration of students' creative solutions, as well as their creative self-efficacy, while originality showed no significant change. Consistent with Urban et al. (2024), who found that ChatGPT can enhance students' creativity, our results indicate that personalization of GAI further amplifies these benefits. The personalized GAI employed in this study served as an AI tutor with anthropomorphic features to enhance social presence (Skulmowski, 2024), providing tailored, goal-oriented feedback and emotional responsiveness throughout the task. Such support likely reduced task uncertainty, strengthened confidence in creative abilities, and fostered greater effort, resulting in more refined and detailed solutions.

Second, eye-tracking results showed that personalized GAI significantly increased gaze transitions between the AI feedback area and the workspace. These cross-AOI transitions reflect cognitive flexibility in understanding, integrating, and monitoring information (Liu and Cui, 2025). Results also showed that creative self-efficacy was positively associated with transition frequency, suggesting that higher self-efficacy may foster more active exploration of AI-generated feedback. Taken together, these findings imply that personalized GAI can boost self-efficacy, amplify exploratory behavior, and encourage frequent shifting between external feedback and internal ideation, thereby facilitating more effective creative thinking.

Finally, exploratory correlations revealed a stable relational structure among creativity dimensions, regardless of GAI type. Eye-tracking data further showed strong correlations in mean fixation duration between AOI 1 and AOI 3 across conditions, indicating a consistent attention allocation pattern during human–GAI collaboration. This co-occurrence of attention to AI feedback and internal idea processing may reflect attentional coupling, the coordinated allocation of attention between two task-relevant foci that integrates their underlying cognitive processes (Richardson et al., 2007). Such coupling may facilitate creative idea generation in GAI-driven collaboration.

6 Conclusion

This within-subjects study used multimodal data to compare the effects of personalized and general-purpose GAI on student creativity. Results showed that personalized GAI significantly improved the quality and elaboration of students' creative solutions, as well as their self-efficacy. Eye-tracking data revealed more transitions between task-relevant AOIs under the personalized condition. The positive correlation between self-efficacy and number of transitions suggests that motivational factors may promote more active information processing. These findings partially support the generative–exploratory model (Finke and Smith, 1992), highlighting a potential interplay between external input and internal ideation during human–GAI co-creativity.

This study also offers practical implications. To maximize the effectiveness of personalized GAI, AI designers should develop cognitively and emotionally adaptive scaffolds, and craft feedback that strategically guides attention. Educators should also provide clear guidance to help students engage meaningfully with AI feedback and integrate it with their own thinking, fostering more effective human–AI co-creativity.

This study has several limitations that should be addressed in future work. First, its controlled setting and small sample may limit ecological validity and generalizability, calling for replication in real classrooms with larger, more diverse participants. Second, it did not assess metacognition, a key factor in GAI-supported creative tasks (Urban et al., 2024). Future research should include metacognitive measures and leverage more multimodal methods to uncover underlying mechanisms. Finally, while personalized GAI improved performance and self-efficacy, its limited effect on originality suggests the embedding of cognitive-based creativity training to foster divergent thinking.

References

Beghetto, R.A., Kaufman, J.C.: Classroom contexts for creativity. High Abil. Stud. **25**(1), 53–69 (2014)

Coskun, A., Cagiltay, K.: A systematic review of eye-tracking-based research on animated multimedia learning. J. Comput. Assist. Learn. **38**(2), 581–598 (2022)

Di Vincenzo, F., Mascia, D., Björk, J., Magnusson, M.: Attention to ideas! Exploring idea survival in internal crowdsourcing. Eur. J. Innov. Manag. **24**(2), 213–234 (2021)

Finke, R.A., Ward, T.B., Smith, S.M.: Creative cognition: Theory, research, and applications. MIT Press (1996)

Guettala, M., Bourekkache, S., Kazar, O., Harous, S.: Generative artificial intelligence in education: advancing adaptive and personalized learning. Acta Informatica Pragensia **13**(3), 460–489 (2024)

Habib, S., Vogel, T., Anli, X., Thorne, E.: How does generative artificial intelligence impact student creativity? Journal of Creativity **34**(1), 100072 (2024)

Hattie, J., Timperleyttie, H.: The power of feedback. Tijdschrift voor Medisch Onderwijs **27**(1), 50–51 (2008). https://doi.org/10.1007/BF03078234

Hernández-Torrano, D., Ibrayeva, L.: Creativity and education: a bibliometric mapping of the research literature (1975–2019). Thinking Skills and Creativity **35**, 100625 (2020)

Joksimovic, S., Ifenthaler, D., Marrone, R., De Laat, M., Siemens, G.: Opportunities of artificial intelligence for supporting complex problem-solving: findings from a scoping review. Comp. Educ. Artif. Intel. **4**, 100138 (2023)

Lee, B.C., Chung, J.: An empirical investigation of the impact of ChatGPT on creativity. Nat. Hum. Behav. **8**(10), 1906–1914 (2024)

Lin, C.F., Yeh, Y.C., Hung, Y.H., Chang, R.I.: Data mining for providing a personalized learning path in creativity: an application of decision trees. Comput. Educ. **68**, 199–210 (2013)

Liu, X., Cui, Y.: Eye tracking technology for examining cognitive processes in education: A systematic review. Computers & Education, 105263 (2025)

Lo, C.K., Hew, K.F., Jong, M.S.Y.: The influence of ChatGPT on student engagement: A systematic review and future research agenda. Computers & Education 105100 (2024)

Maity, S., Deroy, A.: Generative ai and its impact on personalized intelligent tutoring systems (2024). arXiv preprint arXiv:2410.10650

Mogavi, R.H., et al.: ChatGPT in education: a blessing or a curse? a qualitative study exploring early adopters' utilization and perceptions. Comp. Human Behav. Artif. Humans **2**(1), 100027 (2024)

Pi, Z., Yang, Y., Zhao, X., Li, X., Chen, S.: Peers' ideas enhance creativity in collaborative design. Learn. Instr. **98**, 102134 (2025)
Puozzo, I.C., Audrin, C.: Improving self-efficacy and creative self-efficacy to foster creativity and learning in schools. Thinking Skills and Creativity **42**, 100966 (2021)
Reeves, B., Nass, C.: The media equation: How people treat computers, television, and new media like real people and places. Cambridge University Press (1996)
Richardson, D.C., Dale, R., Kirkham, N.Z.: The art of conversation is coordination. Psychol. Sci. **18**(5), 407–413 (2007)
Runco, M.A., Jaeger, G.J.: The standard definition of creativity. Creat. Res. J. **24**(1), 92–96 (2012)
Skulmowski, A.: Placebo or assistant? generative AI between externalization and anthropomorphization. Educ. Psychol. Rev. **36**(2), 1–18 (2024)
Sternberg, R.J., Lubart, T.I.: Defying the crowd: cultivating creativity in a culture of conformity. Free press (1995)
Tierney, P., Farmer, S.M.: Creative self-efficacy: Its potential antecedents and relationship to creative performance. Acad. Manag. J. **45**(6), 1137–1148 (2002)
Torrance, E.P.: Torrance tests of creative thinking. Educational and psychological measurement (1966)
Urban, M., et al.: ChatGPT improves creative problem-solving performance in university students: an experimental study. Comput. Educ. **215**, 105031 (2024)
Vancouver, J.B., Kendall, L.N.: When self-efficacy negatively relates to motivation and performance in a learning context. J. Appl. Psychol. **91**(5), 1146 (2006)
Wei, Y., Shi, Y., MacLeod, J., Yang, H.H.: Exploring the factors that influence college students' academic self-efficacy in blended learning: A study from the personal, interpersonal, and environmental perspectives. SAGE Open **12**(2), 21582440221104816 (2022)
Yeh, Y.: Research and methods. In: Runco, M.A. Pritzker, S.R. (eds.) Encyclopedia of creativity, Vol. 2, pp. 291–298. Academic Press (2011)